

# Research Overview Document (Supervisor Reference)

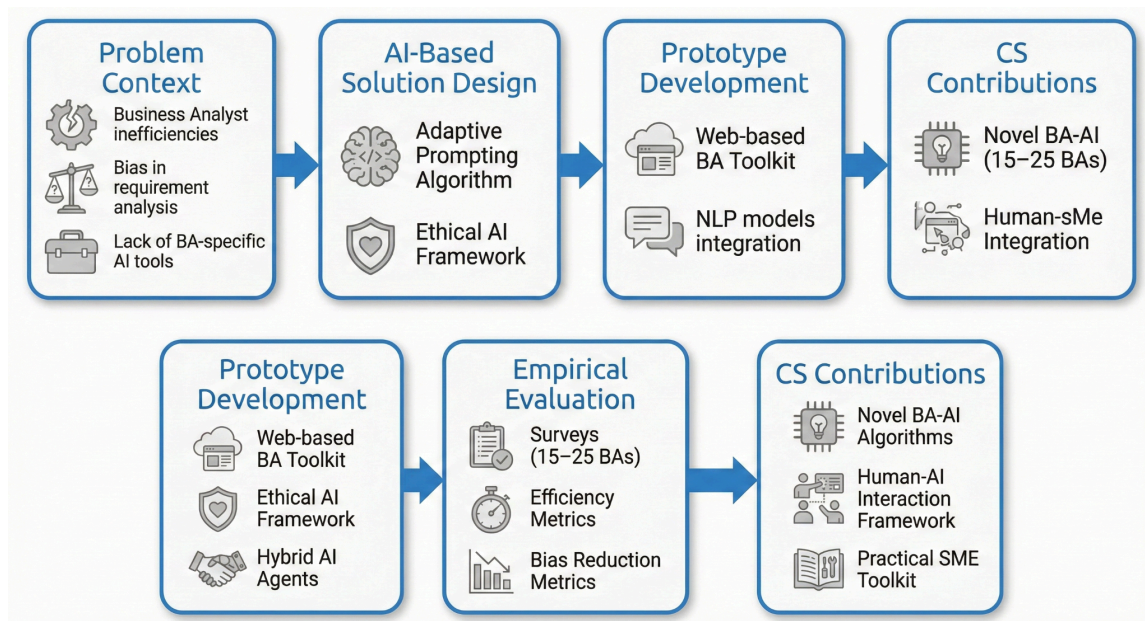
Final Year Research (BCS [Hons])

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## 1. Research Title

AI-Enhanced Methodologies and Agents for Business Analysis in Software Development



## 2. Research Focus

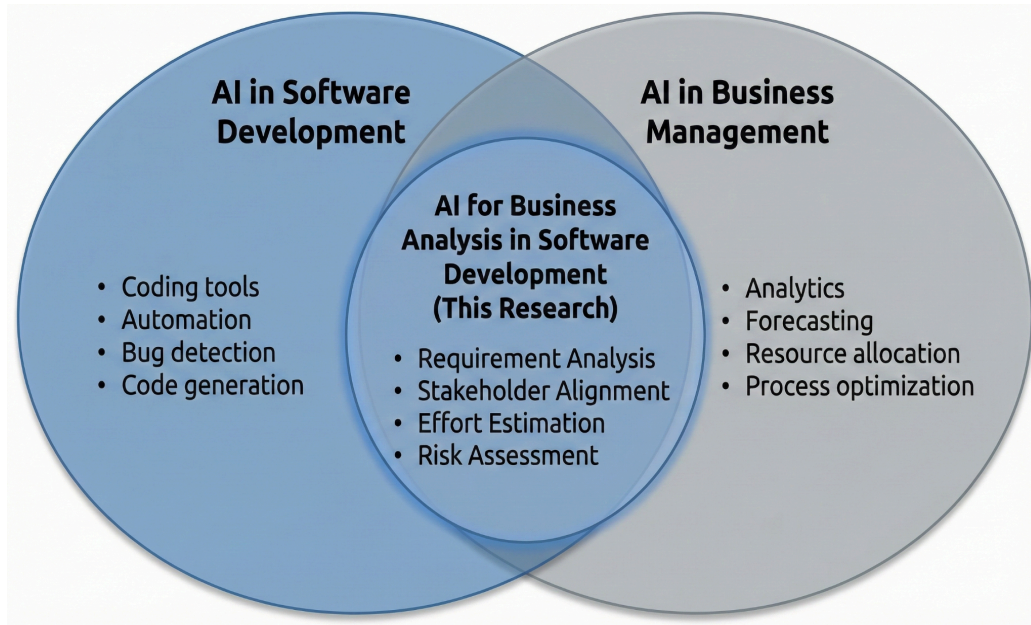
This research explores how Artificial Intelligence (AI) can assist Business Analysts (BAs) in software development projects.

Instead of replacing human analysts, the goal is to:

- Improve efficiency
- Improve accuracy

- Reduce bias
- Support ethical decision-making
- Enhance collaboration between clients and developers

The idea is to create AI-based tools (smart prompts and hybrid AI agents) that help BAs perform their tasks better.

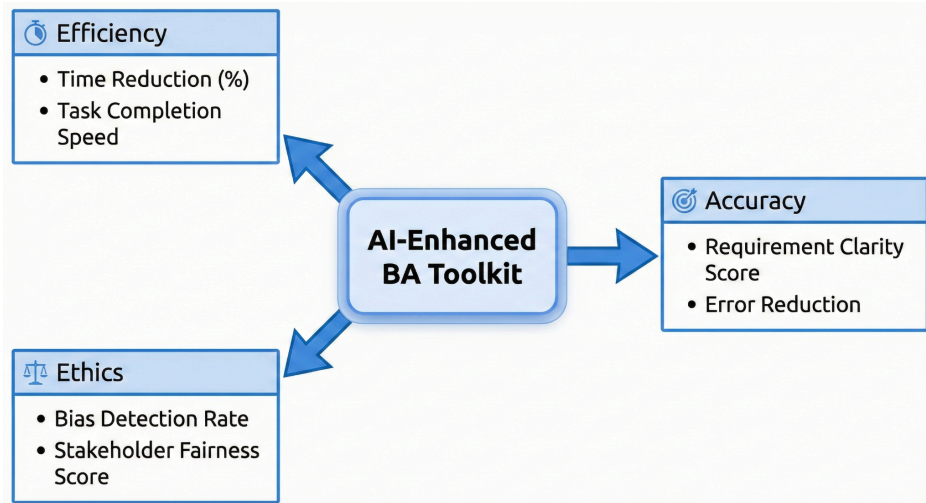


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### 3. Why This Research is Important

- AI adoption is rapidly increasing in software development (AI Boom).
- Many organizations report a lack of AI-related skills.
- Most AI tools focus on coding, not Business Analysis.
- Business Analysts play a critical role in understanding human needs, emotions, culture, and expectations — something AI alone cannot replace.

This research aims to enhance BAs with AI support, creating “AI-augmented professionals” rather than replacing them.



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## 4. Main Research Question

How can AI tools/frameworks improve Business Analyst efficiency, accuracy, and ethical decision-making in software projects?

### Sub-Questions

1. Can adaptive prompting improve requirements gathering?
2. Can ethical AI frameworks reduce bias in analysis?
3. How can hybrid AI agents improve team collaboration?

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## 5. How I Arrived at This Topic

- Background in Computer Science and Software Development
- Internship experience in Business Analysis (at ideahub [pvt] ltd)
- Observed repetitive BA tasks that AI could assist with
- Explored multiple AI-related ideas (e.g., fake news detection, agriculture AI, emotions and mental health assistance, AI for BAs)

- Identified a clear research gap in AI support specifically for Business Analysts
- Noticed relevance to Sri Lanka's growing IT and fintech sectors

This topic aligns with both academic value and real industry needs. (IIBA - International Institute of Business Analyst)

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## 6. Literature Skim Summary

### Search Sources

- Google Scholar (+Labs) : <https://scholar.google.com>
- IEEE Xplore : <https://ieeexplore.ieee.org>
- arXiv : <https://arxiv.org>
- Industry Reports (PwC, Deloitte, etc.) : <https://www.deloitte.com> and <https://www.pwc.com>

### Keywords Used

- AI-enhanced business analysis
- Hybrid AI agents in software development
- AI prompt engineering for requirements engineering
- AI in SDLC (2023–2026)

### Key Observations

- Many studies focus on AI for coding (e.g., GitHub Copilot).
- Few studies address AI support for Business Analysts.
- No recent research specifically targets:
  - Adaptive prompting for BA workflows



- Ethical AI frameworks for requirement analysis
- Hybrid AI agents tailored for SMEs

## Research Papers

- Hybrid Methodologies in Software Development with AI: The Role of Business Analysis in Economic Optimization of Personalized Medicine <sup>(Nov.2025)</sup>  
(Link:[https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=5579173](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5579173))
  - AI-driven business analytics and decision making (Aug.2024)  
(Link:<https://wjarr.com/sites/default/files/WJARR-2024-3093.pdf>)
  - Generative AI in Software Engineering <sup>(Oct.2025)</sup>  
(Link:[https://www.dfki.de/fileadmin/user\\_upload/DFKI/Medien/News/2025/Wissenschaftliche\\_Exzellenz/Generative\\_AI\\_in\\_Software\\_Engineering\\_Transforming\\_the\\_Software\\_Development\\_Processes\\_2025.pdf](https://www.dfki.de/fileadmin/user_upload/DFKI/Medien/News/2025/Wissenschaftliche_Exzellenz/Generative_AI_in_Software_Engineering_Transforming_the_Software_Development_Processes_2025.pdf))
  - Integrating Generative AI into the Software Development Lifecycle: Impacts on Code Quality and Maintenance <sup>(Aug.2024)</sup>  
(Link:<https://ijsra.net/sites/default/files/IJSRA-2024-1837.pdf>)
  - Integrating Generative AI for Advancing Agile Software Development and Mitigating Project Management Challenges <sup>(2024)</sup>  
(Link:[https://thesai.org/Downloads/Volume15No3/Paper\\_6-Integrating\\_Generative\\_AI\\_for\\_Advancing\\_Agile\\_Software.pdf](https://thesai.org/Downloads/Volume15No3/Paper_6-Integrating_Generative_AI_for_Advancing_Agile_Software.pdf))
  - The Business Analyst Guide (AI and Problems)  
(Link:[https://www.itworx.com/wp-content/uploads/2025/03/The-Worxian-Wizard\\_Software-Business-Analysis-1.pdf](https://www.itworx.com/wp-content/uploads/2025/03/The-Worxian-Wizard_Software-Business-Analysis-1.pdf))
  - Artificial Intelligence Applications in Lean Startup Methodology: A Bibliometric Analysis of Research Trends and Future Directions <sup>(Jun.2025)</sup>  
(Link:<https://www.arxiv.org/pdf/2512.22164>)
  - AI-Driven Business Intelligence: Unlocking the Future of Decision-Making <sup>(Sep.2023)</sup>  
(Link:<https://www.espijournals.org/IJACT/2023/Volume1-Issue2/IJACT-V1I2P108.pdf>)
  - Integrating AI to Enhance Business Analysis Education <sup>(Dec.2024)</sup>  
(Link:<https://koreascience.kr/article/JAKO202405656880175.page>)
  - AI-enhanced Business Intelligence for decision-making <sup>(2025)</sup>  
(Link:<https://www.sciencedirect.com/science/article/pii/S1877050925028303>)
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## 7. Identified Research Gap

Current AI tools:

- Focus mainly on developers.
- Do not address human-centric requirement elicitation.
- Lack BA-specific algorithms.

- Ignore ethical prompting and bias reduction.

There is no recent (2024–2026) focused research on AI methodologies specifically designed for Business Analysts.

This research fills that gap.

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## 8. Methodology

### Research Approach

- Mixed Methods Approach
  - Design Science (build prototypes)
  - Empirical Evaluation (surveys, interviews, experiments)

### Research Flow (6–8 Months Plan)

#### Month 1

- Detailed literature review
- Finalize research questions

#### Month 2

- Design AI methodologies
- Develop adaptive prompting algorithms

#### Months 3–4

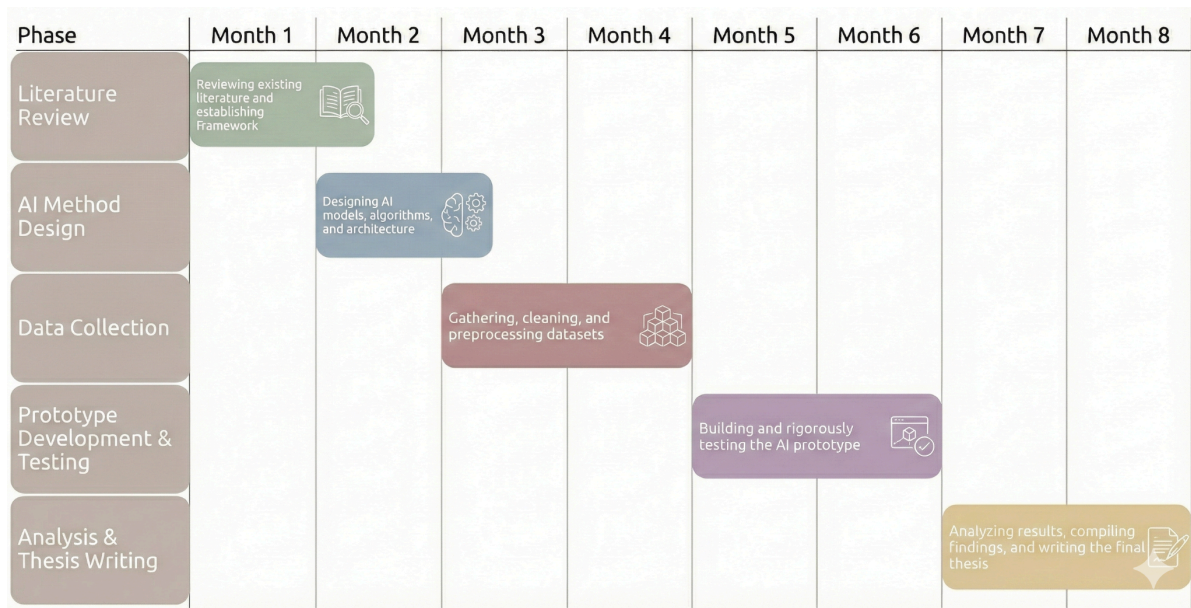
- Collect data from 15–25 Sri Lankan Business Analysts
- Conduct interviews and surveys

#### Months 5–6

- Develop prototype toolkit (web-based)
- Test performance improvements

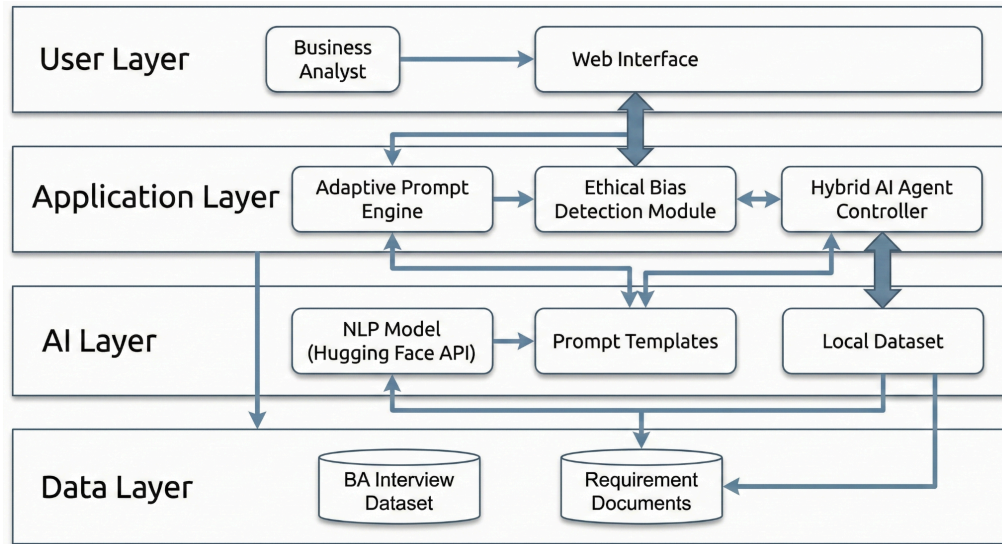
#### Months 7-8

- Analyze results
- Write thesis
- Prepare conference paper



## 9. Expected Deliverables

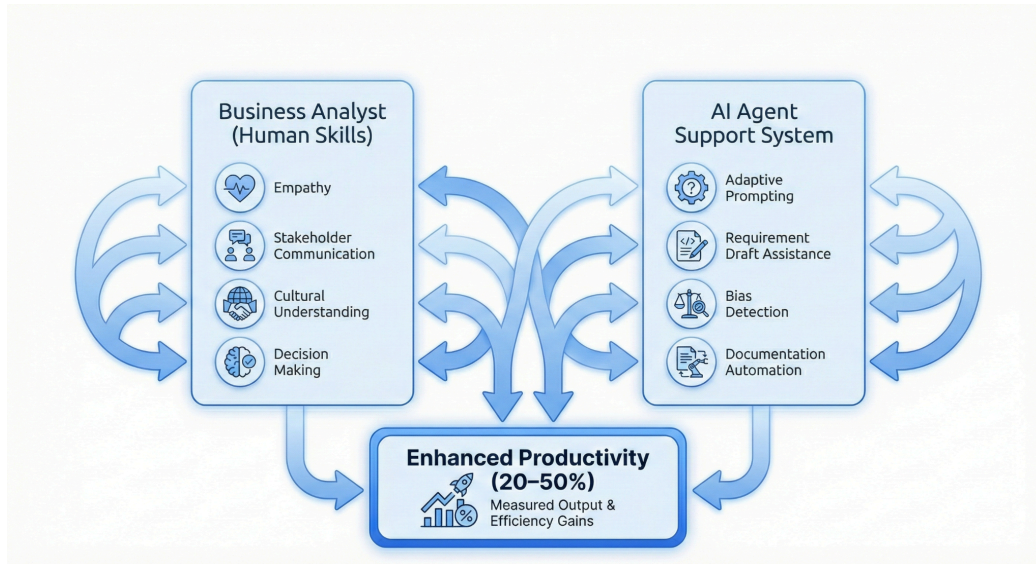
- Prototype BA AI Toolkit (Web Application)
- Custom anonymized dataset (if possible open-source)
- Final thesis report
- GitHub repository with code
- Conference paper submission (IEEE/ACM preferred)



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## 10. Expected Outcomes

- 20–50% efficiency improvement in BA tasks
- Reduced bias in requirement documentation
- Clear insights into AI limitations
- Practical framework for SMEs in Sri Lanka



## 11. Risks & Mitigation

| Risk                     | Mitigation                     |
|--------------------------|--------------------------------|
| Limited data access      | Use simulation datasets        |
| Low survey participation | Use professional networks      |
| AI model limitations     | Use hybrid human-AI evaluation |

## 12. Computer Science Contributions

- Novel adaptive prompting algorithm for BA tasks
- Human-AI interaction framework for Software Engineering
- Ethical AI integration in Requirements Engineering
- Scalable toolkit adaptable to Agile workflows (e.g., Jira integration)
- Strengthening AI-Business Analysis synergy research area



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## 13. Broader Impact

- Supports Sri Lanka's growing IT ecosystem
- Bridges academia and industry practice
- Enhances Business Analyst roles rather than replacing them
- Builds AI awareness in SMEs

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## 14. Personal Commitment

I am committed to completing this research with:

- Strong academic rigor
- Ethical data handling
- Practical industry relevance
- Publication-quality output

I am open to refining and improving any part of this proposal based on your guidance.

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